

Effective Spectral Dimension and the Banach Fixed-Point Theorem in Scale-Dependent Geometric Field Theory

David A. Besemer^{1,*}

¹*Independent Researcher*

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We derive the effective spectral dimension D^* of spacetime within the framework of Scale-Dependent Geometric Field Theory (SDGFT), a zero-parameter approach in which all physical constants emerge from two topological axioms: the Fibonacci-lattice conflict $\Delta = 5/24$ and the elementary lattice tension $\delta = 1/24$, both rooted in the unique self-dual regular four-dimensional polytope, the 24-cell $\{3, 4, 3\}$.

We present two independent derivations. First, a *tree-level algebraic* route yields the exact rational value $D_{\text{tree}}^* = 67/24$. Second, a *self-referential fixed-point* iteration defines a continuous flow operator $\mathcal{T}(D) = \Delta^{-1/D} \varphi \Delta^{\Delta\delta}$, whose unique attractor we prove to be a *Banach contraction mapping* on the interval $I = [2.4, 3.6]$. The analytical derivative $\mathcal{T}'(D) = \mathcal{T}(D) \ln \Delta/D^2$ satisfies the Lipschitz bound $|\mathcal{T}'(D)| < 0.836 < 1$ uniformly on I , guaranteeing existence and uniqueness of the fixed point $D_{\text{fp}}^* \approx 2.79676$. Numerical iteration from any starting value in I converges to machine precision (10^{-15}) in exactly 60 steps.

The resulting $f(R)$ gravity exponent $n = D^*/2 = 67/48$ determines all downstream gravitational, inflationary, and cosmological observables of the theory without free parameters.

I. INTRODUCTION

The problem of determining the effective dimensionality of spacetime at short distances has attracted sustained attention from multiple approaches to quantum gravity, including causal dynamical triangulations [1], asymptotic safety [2, 3], loop quantum gravity [4], and Hořava–Lifshitz gravity [5]. A common theme is the observation of a *spectral dimension* that decreases from 4 in the infrared (IR) to approximately 2 in the ultraviolet (UV), suggesting that the microstructure of spacetime is fundamentally lower-dimensional.

Scale-Dependent Geometric Field Theory (SDGFT) [6–8] takes a more radical stance: spacetime dimensionality is not merely an observable but a *derived quantity*, uniquely fixed by two rational topological axioms associated with the 24-cell polytope $\{3, 4, 3\}$ —the unique self-dual regular polytope in four dimensions. The two axioms are:

$$\Delta = \frac{5}{24}, \quad (1)$$

$$\delta = \frac{1}{24}. \quad (2)$$

Here Δ encodes the Fibonacci-lattice conflict ($F_5 = 5$ Fibonacci chains distributed over the 24 vertices), while δ represents the elementary lattice tension ($1/V_{24}$, the inverse vertex count).

From these axioms alone, three emergent quantities follow algebraically:

1. *Golden ratio*: $\Delta/\delta = 5 \implies \varphi = \frac{1+\sqrt{5}}{2}$.

2. *Cone half-angle*: $\Delta + \delta = \frac{1}{4} = \sin^2 30^\circ \implies \theta_{\text{max}} = 30^\circ$.

3. *Six-cone partition*: $360^\circ / (2 \times 30^\circ) = 6$ congruent cones on S^3 .

The central quantity of the present paper is the effective spectral dimension D^* . We derive it by two independent routes: a tree-level algebraic formula (Sec. II) and a self-referential fixed-point iteration (Sec. III). The fixed-point route is mathematically richer and constitutes the primary result: a rigorous proof, via the Banach contraction mapping theorem, that the operator $\mathcal{T}: I \rightarrow I$ possesses a *unique, globally attracting* fixed point.

II. TREE-LEVEL DERIVATION

At tree level, the effective dimension is obtained by correcting the topological dimension $d = 3$ of the spatial S^3 base manifold by two competing effects: the cone partition removes $\sin^2 \theta_{\text{max}} = \frac{1}{4}$ of the angular volume, while the lattice tension δ restores a quantum correction. The result is:

$$D_{\text{tree}}^* = 3 - \sin^2 30^\circ + \delta = 3 - \frac{1}{4} + \frac{1}{24} = \frac{72 - 6 + 1}{24} = \frac{67}{24}. \quad (3)$$

This can equivalently be written as

$$D_{\text{tree}}^* = 3 - \Delta = 3 - \frac{5}{24} = \frac{67}{24}, \quad (4)$$

since $\Delta = \sin^2 30^\circ - \delta = \frac{1}{4} - \frac{1}{24} = \frac{5}{24}$. Both forms are verified as exact rational identities in the SDGFT codebase [9].

The $f(R)$ gravity exponent is

$$n_{\text{tree}} = \frac{D_{\text{tree}}^*}{2} = \frac{67}{48} \approx 1.39583, \quad (5)$$

* david.besemer@sdgft.org

and the combination $2n - 1 = 43/24$ appears throughout inflationary slow-roll formulae [12].

III. SELF-REFERENTIAL FIXED-POINT DERIVATION

A. The flow operator

The tree-level derivation treats the lattice correction as a first-order perturbation. A fully self-consistent treatment must account for the fact that the spectral dimension itself enters the exponent controlling the lattice deformation. This leads to the self-referential flow operator

$$\boxed{\mathcal{T}(D) = \Delta^{-1/D} \varphi \Delta^{\Delta\delta}}. \quad (6)$$

The three factors have the following physical interpretation:

- $\Delta^{-1/D}$: the scale-dependent deformation of the lattice, with the exponent $-1/D$ encoding the feedback of the effective dimension on the deformation amplitude.
- φ : the golden-ratio scaling, reflecting the Fibonacci self-similarity of the 24-cell decomposition.
- $\Delta^{\Delta\delta}$: a topological quantum correction of magnitude $\Delta^{5/576}$, representing the interplay between the two fundamental lattice scales.

A fixed point D^* of $\mathcal{T}$ satisfies

$$D^* = \mathcal{T}(D^*) = \Delta^{-1/D^*} \varphi \Delta^{\Delta\delta}. \quad (7)$$

This is a transcendental equation that cannot be solved in closed form; however, we shall prove that the iteration $D_{k+1} = \mathcal{T}(D_k)$ converges to a unique fixed point for *any* starting value in a suitable interval.

B. Analytical derivative

The derivative of $\mathcal{T}(D)$ with respect to D is computed via the chain rule applied to the D -dependent factor $\Delta^{-1/D} = \exp(-\ln \Delta / D)$:

$$\mathcal{T}'(D) = \mathcal{T}(D) \frac{\ln \Delta}{D^2}. \quad (8)$$

Since $\Delta = 5/24 < 1$, we have $\ln \Delta < 0$, and therefore $\mathcal{T}'(D) < 0$ for all $D > 0$: the flow operator is *monotonically decreasing*. This guarantees that the iteration alternates around the fixed point, producing the characteristic oscillatory convergence seen in the numerical data (Table I).

At the fixed point itself:

$$\mathcal{T}'(D_{\text{fp}}^*) = D_{\text{fp}}^* \cdot \frac{\ln(5/24)}{(D_{\text{fp}}^*)^2} = \frac{\ln(5/24)}{D_{\text{fp}}^*} = \frac{-1.56862}{2.79676} \approx -0.5609 \quad (9)$$

IV. BANACH CONTRACTION MAPPING PROOF

Theorem 1 (Existence and uniqueness of D^*). *Let $I = [2.4, 3.6]$ and let $\mathcal{T}: I \rightarrow \mathbb{R}$ be defined by Eq. (6). Then:*

- (i) $\mathcal{T}(I) \subseteq I$ (self-mapping),
- (ii) $\sup_{D \in I} |\mathcal{T}'(D)| \leq q < 1$ with $q = 0.836$ (contraction),
- (iii) $\mathcal{T}$ possesses a unique fixed point $D_{\text{fp}}^* \in I$, and the Picard iteration $D_{k+1} = \mathcal{T}(D_k)$ converges to D_{fp}^* for any $D_0 \in I$.

Proof. (i) **Self-mapping.** $\mathcal{T}$ is continuous on $(0, \infty)$. Since $\Delta = 5/24 < 1$ and the exponent $-1/D$ is negative for $D > 0$, the factor $\Delta^{-1/D}$ is a decreasing function of D that satisfies $\Delta^{-1/D} > 1$. Direct evaluation at the boundary points yields:

$$\mathcal{T}(2.4) = \left(\frac{5}{24}\right)^{-1/2.4} \cdot \varphi \cdot \left(\frac{5}{24}\right)^{5/576} \approx 3.246, \quad (10)$$

$$\mathcal{T}(3.6) \approx 2.556. \quad (11)$$

Since $\mathcal{T}$ is continuous and monotonically decreasing on I , and $2.4 \leq 2.556 \leq \mathcal{T}(D) \leq 3.246 \leq 3.6$ for all $D \in I$, we conclude $\mathcal{T}(I) \subseteq I$.

(ii) **Contraction.** By Eq. (8), $|\mathcal{T}'(D)|$ is a decreasing function of D on I (since both $\mathcal{T}(D)$ and $1/D^2$ decrease with D). The supremum is therefore attained at the left endpoint:

$$\sup_{D \in I} |\mathcal{T}'(D)| = |\mathcal{T}'(2.4)| \approx 0.836 =: q < 1. \quad (12)$$

This is verified numerically at 1001 uniformly spaced sample points across I (see Table II).

(iii) **Banach fixed-point theorem.** Since $(I, |\cdot|)$ is a complete metric space, $\mathcal{T}(I) \subseteq I$, and $\mathcal{T}$ is a contraction with Lipschitz constant $q < 1$, the Banach fixed-point theorem guarantees existence and uniqueness of the fixed point $D_{\text{fp}}^* \in I$, and convergence of the Picard iteration from any starting point in I . The a priori error bound is

$$|D_k - D_{\text{fp}}^*| \leq \frac{q^k}{1 - q} |D_1 - D_0|. \quad (13)$$

With $q = 0.836$ and $|D_1 - D_0| \approx 0.31$ (for $D_0 = 3.0$), this gives a bound of 10^{-15} after $k \approx 190$ iterations. In practice, the effective contraction constant near the fixed point is $|\mathcal{T}'(D_{\text{fp}}^*)| \approx 0.561$, leading to the a posteriori bound

$$|D_k - D_{\text{fp}}^*| \lesssim 0.561^k \times 0.31, \quad (14)$$

which predicts convergence to 10^{-15} in $k \approx \ln(10^{-15}/0.31)/\ln(0.561) \approx 57$ iterations, consistent with the observed 60 steps. $\square$

V. NUMERICAL RESULTS

A. Fixed-point value

Starting from $D_0 = 3.0$, the Picard iteration converges in exactly 60 steps to

$$D_{\text{fp}}^* = 2.796\,764\,908\,336\,852, \quad (15)$$

satisfying $|\mathcal{T}(D_{\text{fp}}^*) - D_{\text{fp}}^*| < 10^{-15}$. The corresponding $f(R)$ exponent is

$$n_{\text{fp}} = \frac{D_{\text{fp}}^*}{2} = 1.398\,382\,454\,168\,426. \quad (16)$$

Note the close agreement with the tree-level value $n_{\text{tree}} = 67/48 \approx 1.39583$; the two differ by only 0.18%.

B. Convergence history

Table I documents the convergence trajectory, showing the characteristic oscillatory approach and the exponential decay of the error $|D_k - D_{\text{fp}}^*|$.

TABLE I. Convergence of the Picard iteration $D_{k+1} = \mathcal{T}(D_k)$ from $D_0 = 3.0$ to the fixed point $D_{\text{fp}}^* = 2.796764908336852$. The iteration exhibits oscillatory convergence with an effective contraction ratio ≈ 0.56 .

k	D_k	$ D_k - D_{\text{fp}}^* $
0	3.000 000 000 000 000	2.03×10^{-1}
1	2.692 492 488 011 853	1.04×10^{-1}
2	2.858 177 400 072 536	6.14×10^{-2}
3	2.763 262 971 810 615	3.35×10^{-2}
5	2.786 154 695 425 669	1.06×10^{-2}
10	2.797 355 729 559 809	5.91×10^{-4}
20	2.796 766 727 985 851	1.82×10^{-6}
30	2.796 764 913 942 087	5.61×10^{-9}
40	2.796 764 908 354 118	1.73×10^{-11}
50	2.796 764 908 336 904	5.28×10^{-14}
55	2.796 764 908 336 848	3.55×10^{-15}
60	2.796 764 908 336 852	0

C. Lipschitz bound across the interval

D. Universality of convergence

The Banach theorem guarantees convergence from *any* starting value $D_0 \in I$. We have verified this numerically for $D_0 \in \{1.5, 2.0, 3.0, 5.0, 10.0\}$; all converge to the same D_{fp}^* within 10^{-10} . In fact, the basin of attraction extends well beyond I , covering $(0, \infty)$, since $\mathcal{T}(D) \in I$ for any $D > 0$ (by the boundedness of $\Delta^{-1/D}$ for $\Delta \in (0, 1)$).

TABLE II. Values of the Lipschitz bound $|\mathcal{T}'(D)|$ at representative points across the interval $I = [2.4, 3.6]$. The supremum $q \approx 0.836$ is attained at the left boundary.

D	$ \mathcal{T}'(D) $
2.4	0.8356
2.6	0.6771
2.8	0.5592
D_{fp}^*	0.5609
3.0	0.4693
3.2	0.3992
3.4	0.3436
3.6	0.2987

VI. PHYSICAL IMPLICATIONS

A. The $f(R)$ gravity action

The derived effective dimension determines the gravitational action of SDGFT:

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R^{67/48}. \quad (17)$$

This is a specific member of the $f(R) = R^n$ family of modified gravity theories [13], with the exponent $n = 67/48$ rigidly fixed by the 24-cell topology rather than fitted to observations.

The Bellini–Sawicki Horndeski parameters [10, 14] follow algebraically:

$$\alpha_{\text{M}} = \frac{n-1}{2n-1} = \frac{19}{86} \approx 0.2209, \quad (18)$$

$$\alpha_{\text{B}} = -\frac{\alpha_{\text{M}}}{2} = -\frac{19}{172} \approx -0.1105, \quad (19)$$

$$\alpha_{\text{T}} = 0, \quad (20)$$

$$\alpha_{\text{K}} = 0. \quad (21)$$

The prediction $\alpha_{\text{T}} = 0$ is non-trivially consistent with the GW170817 constraint $|c_T/c - 1| < 10^{-15}$ [18].

B. Inflationary observables

Using the Mukhanov–Sasaki formulation for R^n inflation in the Jordan frame [12, 15], the slow-roll parameters determine:

$$n_s = 1 - \frac{2(2n-1)}{N_e(2n-1) + n} \approx 0.9671, \quad (22)$$

$$r = \frac{48(2n-1)^2}{[N_e(2n-1) + n]^2} \approx 0.0130, \quad (23)$$

where $N_e \approx 59.95$ e-folds. These predictions are consistent with Planck 2018 ($n_s = 0.9649 \pm 0.0042$) [16] and BICEP/Keck ($r < 0.036$ at 95% CL) [17].

C. Spectral dimension in the UV

The flow operator $\mathcal{T}$ also encodes the UV behavior: $\mathcal{T}(D) \rightarrow 2$ as $D \rightarrow 2^+$, consistent with the universal $d_s \rightarrow 2$ UV limit observed in causal dynamical triangulations and asymptotic safety. The full spectral dimension profile—from 2 in the UV to $D_{\text{fp}}^* \approx 2.80$ in the IR—will be presented in [11].

VII. DISCUSSION

The key result of this paper is the rigorous establishment of a *unique, globally attracting* fixed point for the SDGFT dimensional flow. Three aspects deserve emphasis:

Mathematical rigor. The proof via the Banach contraction mapping theorem is constructive: it not only guarantees existence and uniqueness but provides explicit error bounds. The effective contraction constant $|\mathcal{T}'(D_{\text{fp}}^*)| \approx 0.561$ indicates strong contraction, explaining the rapid convergence (60 steps to machine precision).

Independence of starting value. The attractor nature of D_{fp}^* means that the effective dimension is not an input to the theory but an *output*, independent of initial conditions. This is a concrete realization of the “no free parameters” philosophy of SDGFT.

Falsifiability. The rigidly fixed exponent $n = D^*/2$ generates testable predictions. The most immediate are

the inflationary observables $n_s \approx 0.967$ and $r \approx 0.013$. The latter is within reach of forthcoming CMB polarization experiments including LiteBIRD [19] and CMB-S4 [20], and a measurement of r inconsistent with 0.013 would falsify the theory.

VIII. CONCLUSIONS

We have derived the effective spectral dimension of spacetime in SDGFT by two independent routes: a tree-level algebraic calculation yielding $D_{\text{tree}}^* = 67/24$, and a self-referential fixed-point iteration converging to $D_{\text{fp}}^* \approx 2.79676$. The fixed-point iteration is rigorously established as a Banach contraction mapping on $I = [2.4, 3.6]$, with Lipschitz constant $q \approx 0.836$ and effective contraction $|\mathcal{T}'(D_{\text{fp}}^*)| \approx 0.561$.

The derived dimension uniquely determines the $f(R) = R^{67/48}$ gravitational action, the Horndeski parameters, and all inflationary observables of the theory—all without adjustable parameters. The resulting predictions are consistent with current observational data and will be decisively tested by next-generation CMB experiments.

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